



Summary

Firmware engineer with experience on XPON/L2/L3/OLT networking equipment and published IEEE research on FPGA acceleration for ML. M.Sc. in Computer and Telematics Engineering focused on low-level and real-time systems.

Professional Experience

- Firmware Engineer, Dellent

2025 Aug – Present | Aveiro, Portugal

Firmware development for XPON with L2, L3 and OLT Switches @AlticeLabs
 - Took ownership of a ~**35K LOC** undocumented **C firmware stack**, reverse-engineering and documenting **3 core modules** to close team knowledge gaps and accelerate onboarding
 - Eliminated ~**5K lines** of redundant code through systematic **refactoring**, following **DRY** principles
 - Integrated Clang formatting, static analysis, and unit tests into the **GitHub Actions CI pipeline** and streamlined compile scripts for better maintainability
 - Produced structured technical documentation for previously undocumented firmware components
- Researcher, Instituto de Telecomunicações

2024 Feb – 2025 Jul | Aveiro, Portugal

- Published **2 first-author IEEE papers** (NorCAS 2025, ICECS 2025) and a co-authored Springer paper (ICCSA 2025) on **FPGA-based DNN acceleration** and ML classification
 - Developed a code generation framework for DNN deployment on FPGAs via Vitis HLS, optimizing latency, resource utilization, and real-time inference at the edge

Publications

- Using Automatic Code Generation to Streamline the Real-Time FPGA Deployment of DNNs at the Edge

IEEE ICECS 2025

2025 Nov

Developed an open-source HLS-oriented code generator in C for rapid deployment of custom DNN architectures on FPGAs targeting beyond 5G scenarios.
- FPGA Acceleration of CNNs at the Edge: A Comparative Study on High-Level Synthesis Frameworks

IEEE NorCAS 2025

2025 Oct

Compared HDL Coder and Vitis HLS across latency, resource usage, and development effort for real-time edge ML deployment on FPGA.
- Understanding Signal Feature Impact in CNN-Based Modulation Classification with Realistic Datasets

ICCSA 2025

2025 Jul

Education

- M.Sc. in Computer and Telematics Engineering, Universidade de Aveiro

2023 Sep – 2025 Jul | Aveiro, Portugal

- **Thesis:** Code generation framework for DNN deployment on FPGAs via High-Level Synthesis
 - Implemented DNN operations in C for single-cycle II in Vitis HLS, including transcendental activation functions
 - Built a parallel code generator that translates GUI input into HLS-ready C files
 - Designed sample-by-sample inference architecture with local memory reuse, enabling beyond-5G real-time scenarios
- B.Sc. Computer and Telematics Engineering, Universidade de Aveiro

2020 Oct – 2023 Jul | Aveiro, Portugal

Skills

Languages C, C++, Java, Python, Bash, VHDL, MIPS Assembly	Hardware & Architecture Computer Architecture, Digital Systems, FPGA, Vitis HLS, RISC-V	Performance Concurrent and Object Oriented Programming, Memory Management
CI & Tooling: Git, GitHub Actions, Clang	Embedded & Real-Time Systems FreeRTOS, Zephyr RTOS, Linux, Peripheral Inspection	Parallel Computing CUDA, OpenMP

Projects

- CSV-Parser

A Fully Deterministic, High-Performance CSV Parser in C

2026 Apr

- Architected 6 parser variants (computed-goto FSM for branchless processors, branch-based, LUT-based) × (scalar, SIMD) as a performance case study in control-flow and memory strategies
 - Used per-worker arena allocators for zero-allocation hot-path memory management

Languages

Portuguese (C2) | English (C2)